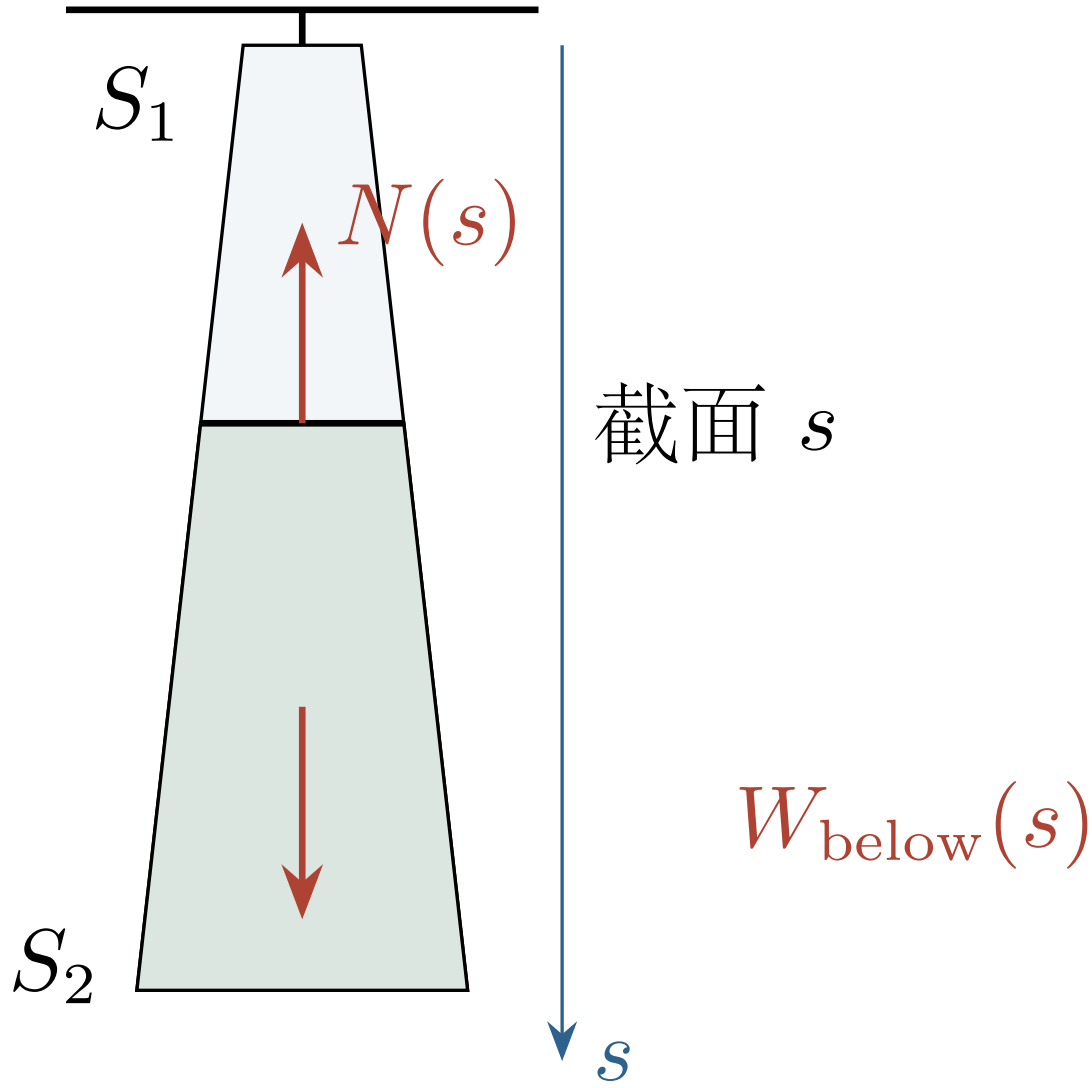

$$W_{\text{below}}(s) = \rho g \int_s^{l_0} S(\xi) d\xi$$

自由端 $N(l_0) = 0$；截面拉力等于阴影段重量